

Frank — AMC Medal Drill L2 / L3 stretch

25 Jun 2026 | ANSWER KEY, method & the trap — for the parent

- 1. [L2] Answer: 168** [cross-number / digit-property · real 2025 Q26 (M2) + hidden condition · difficulty 5 vs real ceiling 5 · 7 marks]

What is the smallest three-digit number with all of these properties: divisible by 3; contains the digit 6; has no digit 0; has all different digits; has a digit sum divisible by 5?

★ Method: Apply the strong filters first (divisibility, no-zero, distinct), then check the SUBTLE last condition before you commit — don't stop at the first multiple you find.

The trap: a solver who skips “has a digit sum divisible by 5” stops at 126 — the planted decoy.

- 2. [L2] Answer: 258** [cross-number / digit-property · real 2025 Q26 (M2) + hidden condition · difficulty 5 vs real ceiling 5 · 7 marks]

What is the smallest three-digit number with all of these properties: divisible by 6; contains the digit 5; has no digit 0; has all different digits; has exactly 1 odd digit?

★ Method: Apply the strong filters first (divisibility, no-zero, distinct), then check the SUBTLE last condition before you commit — don't stop at the first multiple you find.

The trap: a solver who skips “has exactly 1 odd digit” stops at 156 — the planted decoy.

- 3. [L3] Answer: 384** [organised counting (residue classes) · scaled twin of 2013 Q27 (M5) · difficulty 20 vs real ceiling 12 · 7 marks]

In how many ways can you choose 3 different numbers from 3 to 22 so that their sum is a multiple of 3?

★ Method: Bucket all numbers by remainder mod 3; the valid triples are 'all-from-one-bucket' + the matching cross-bucket combinations.

Why it's L3: $C(20,3) = 1140$ triples — impossible to list. Sort the numbers into remainder buckets mod 3 and count the bucket combinations.

- 4. [L3] Answer: 506** [organised counting (residue classes) · scaled twin of 2013 Q27 (M5) · difficulty 24 vs real ceiling 12 · 7 marks]

In how many ways can you choose 3 different numbers from 3 to 26 so that their sum is a multiple of 4?

★ Method: Bucket all numbers by remainder mod 4; the valid triples are 'all-from-one-bucket' + the matching cross-bucket combinations.

Why it's L3: $C(24,3) = 2024$ triples — impossible to list. Sort the numbers into remainder buckets mod 4 and count the bucket combinations.

- 5. [L3] Answer: 4** [cross-number / digit-property · scaled twin of 2025 Q26 (M2) · difficulty 6 vs real ceiling 5 · 8 marks]

How many four-digit numbers have all of these properties: divisible by 13; has all different digits; has no digit 0; has exactly 3 odd digits; has a digit sum divisible by 3; contains the digit 2?

★ Method: Don't hunt for examples — partition by the binding condition (digit sum / parity) and count each class.

Why it's L3: ~9000 four-digit candidates — you cannot list them. Count structurally: fix the digit-sum/parity buckets and use divisibility.

- 6. [L3] Answer: 48** [organised counting (digit positions) · scaled twin of 2024 Q28 (M5) · difficulty 250 vs real ceiling (n/a) · 8 marks]

You write the 250 numbers 3, 6, 9, ... up to 750. How many times do you write the digit '0'?

★ Method: Go place-column by place-column: count how often the digit appears in the units, then tens, then hundreds across the whole list.

Why it's L3: 250 terms up to 750 — you can't write them out. Count the digit '0' one place-column at a time (units, tens, hundreds).

Every answer is solver-verified (unique). L2 keeps real-paper difficulty but plants a decoy equal to the common slip; L3 is deliberately pushed ABOVE the real-paper ceiling and out of hand-enumeration range. Topics: Frank's 2018 gaps (Q27 enumeration, Q30 cross-number).